

SERGIO FALCON

FULL STACK DEVELOPER

📍 KNOXVILLE, UNITED STATES

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👤 SUMMARY

Software Engineer experienced in building and operating full-stack applications, machine learning systems, and real-time monitoring platforms in production environments. Proficient in Python, Rust, C#, JavaScript, Docker, SQL, and modern software architectures, with hands-on experience developing data pipelines, cloud-hosted services, and computer vision solutions. Passionate about applying software engineering, AI, and data-driven technologies to solve complex technical challenges.

📁 EMPLOYMENT HISTORY

Software Developer at Aisin, Clinton, TN

May 2024 — Present

- Designed and built a modern TPDA system to replace a legacy application, streamlining production workflows and removing reliance on a costly legacy platform
- Engineered a role-based inventory and checkout system managing 43,000+ items across multiple locations using Vue and .NET Core (C#)
- Integrated LIFX smart lights into a synchronized factory-floor system that dynamically displays individual machine statuses (running smoothly, out of order, requires assistance) and alerts the appropriate workers
- Built the DCM Monitor application using FastAPI and Svelte to capture and review die-cast machine imagery, enabling rapid identification of slag and debris overflow conditions while supporting computer vision data collection efforts
- Designed and implemented a data collection and labeling pipeline for a YOLO-based computer vision system, establishing the foundation for automated detection of slag buildup in die-cast machines
- Created a cleanroom sensor monitoring application that displays real-time sensor data on intuitive gauges, enhancing operational oversight using Svelte and Axum (Rust)
- Set up and deployed a blue-green deployment strategy to minimize application downtime during releases

Full Stack Developer at RegScale, Knoxville, TN

November 2022 — November 2023

- Developed on a full stack web application using Angular, .NET Core, C#, Python, GraphQL, SQL, Docker, Bootstrap, and Kendo UI
- Developed and implemented features for document template management, specifically for substituting parameters in a word document, along with creating, saving, updating, and deleting these documents
- Created a user interface for a user to control, monitor, and manage a Job Scheduler for the RegScale CLI
- Led the initiative to implement 508 Web Accessibility features across the application, ensuring compliance with federal standards and enhancing usability
- Increased Web Accessibility coverage to 95%, reducing the number of common accessibility issues by 80%

Full Stack Developer at GovReady, Remote

September 2021 — November 2022

- Developed on a full stack web application using React, Django, DjangoRestFramework, Jinja, Redux, PostgreSQL, Docker, Bootstrap, Material UI, Cypress, and Selene
- Worked alongside a team on the development of a real-time collaboration document editor, integrating a suite of components for templating, polling, and enhanced interactivity
- Developed React components to leverage backend RESTful APIs to populate client data and allow users to manage data as needed
- Implemented systems that adhered to OSCAL standards
- Developed on a full stack web application using React, Django, PostgreSQL, Docker, Cypress, and Material UI
- Created a RESTful API that allowed for seamless integration with other applications
- Developed responsive user interfaces using Material UI that delivered a consistent and intuitive experience across all devices
- Created automated end-to-end tests using Cypress

Software Developer at Climalytics LLC, Austin, TX

May 2019 — September 2019

- Developed on a mobile application using React Native that improved user experience and increased app downloads by 20%
- Created a long-distance climbing competition dashboard to measure a user's current distance climbed and current ranking, utilized across various climbing gyms
- Implemented automated testing with Jest to ensure code quality and reduce the number of bugs in production
- Redesigned the user dashboard to create a user-friendly experience with ease to access a user's session data

📁 PROJECTS

climbtrainer.app - Climbing Training Platform

- Designed and built a full-stack climbing training application that generates personalized training programs and automatically adjusts them based on each user's goals, feedback, and progress

🎓 EDUCATION

Master of Science: Computer Science, University of Texas, San Antonio

August 2021 — May 2023

Bachelor of Science: Computer Science, University of Texas, San Antonio

August 2012 — December 2017

Concentration in Software Engineering and Minor in Information Systems